

statreviewer

Automated Manuscript Analysis



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Outline

- Background: What problem are we trying to solve?
- Overview: What's inside of StatReviewer?
- Report: How to interpret a report?

Background

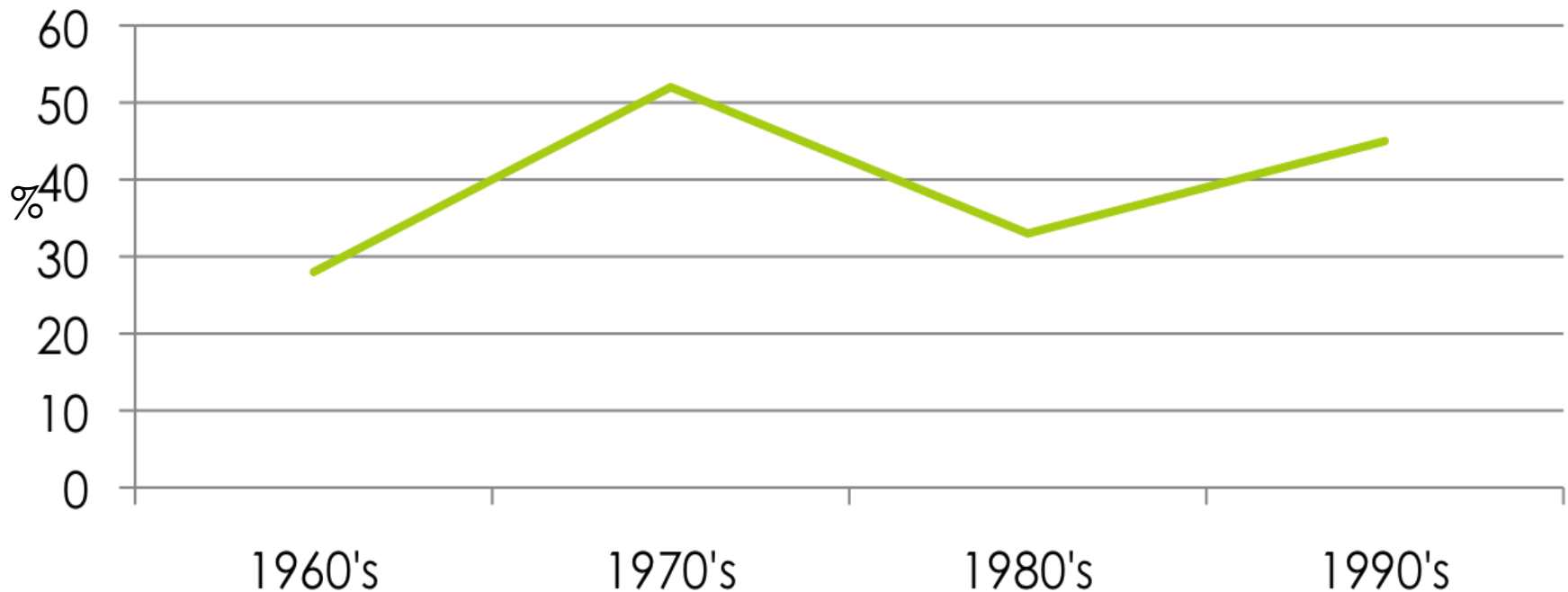
- What problem are we trying to solve?

The Problem with Statistical Reporting in Journals

- Incorrect choice of statistical techniques
- Flawed application of statistical methods
- Omission of crucial information to replicate study
- Erroneous conclusions
- Choice of statistical methods susceptible to bias or exaggerated claims that will fail to be replicated

How big of a problem is poor statistical analyses in medical journals?

Average proportion of “acceptable” papers by decade



Statistics and Peer Review

“...the majority of statistical analyses are performed by people with an inadequate understanding of statistical methods. They are then peer reviewed by people who are generally no more knowledgeable”

-Douglas Altman

- Lack of experts with requisite skills
- Reliance on non-experts – prone to reinforce common errors and misconceptions
- Statistical consultants can be costly (\$75-\$200 per review)
- Consultants often overstretched
 - Rushed reviews
 - Inconsistent application of standards

Overview

- What's inside of StatReviewer?

statreviewer...



Identifies problems with scientific manuscripts



Creates a statistical and methodological issue report



Is instantaneous and fully automated

What's Inside



INGEST

- ✓ Document files or plain-text accepted
- ✓ API-driven
- ✓ Document sections are extracted and prepared for scan
- ✗ For best results, key sections should be present and identifiable using common titles
Such as "Abstract", "Introduction", "Methods"

ANALYZE

Thousands of Checks

A robust arsenal of algorithms are employed to identify everything from large document patterns to small markers

Expert-Informed Artificial Intelligence

We use industry standards as a basis for Machine Learning, not blind unsupervised learning

Always Improving

Continuous feedback loops hone and advance our platform

ANALYZE

What We Find

- ✓ Reporting style (e.g., precision of estimates, confidence intervals, etc.)
- ✓ Error checking (e.g., inferential statistics, proportion checks, proper confidence interval coverage)
- ✓ Inferential statistics (e.g., types of tests, assumptions, reporting of values)
- ✓ Descriptive statistics (e.g., style of reporting, appropriateness of statistic, avoidance of SEM).
- ✓ Ethical approvals (e.g., IRB, ACUC, ethical board approvals, consent)
- ✓ Type-I error control
- ✓ Statistical power analyses
- ✓ Interpretation of results (e.g., misinterpretation of statistical significance, confidence intervals).
- ✓ Many, many more...

ANALYZE

Statistical / Methodological Review

Reporting

What was done?

What was found?

How to interpret?

vs

Design

Are these tests appropriate?

Are the estimates biased or incorrect?

Are the conclusions appropriate?

Report

- How to interpret a report?
- Two sections of report:
 - Fundamentals of reporting (all)
 - Reporting checklist (design-specific, some)

Report Layout

Acrobat Reader File Edit View Window Help

charta-report (2).pdf

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1 / 6 125%

StatReviewer Report

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Issues Found

1

When reporting a proportion in the abstract (e.g., "66%"), it is important to also report the numerator and denominator. For example, 50/100 (50%) patients recovered fully. Providing this information will allow your readers to evaluate the sample size and any missing data.

More Information About This Issue

Sample sizes can vary from outcome to outcome, so reporting the actual sample sizes used to estimate the proportion can help eliminate uncertainty. Reporting the numerator/denominator (%) such as "25/100 (25%)" provides substantial information about the the frequency counts used to estimate the proportion.

Examples

This text: "40% of patients responded to the survey".
Can be revised to: "230 of 575 (40%) of patients responded to the survey".

This text: "The success rate was 22% of cells".
Can be revised to: "Out of 309 cells, 68 (22%) were successful".

This text: "33% of the placebo group and 55% of the active treatment..."
Can be revised to: "33/100 (33%) of the placebo group and 550/1000 (55%) of the active treatment..."

2

The abstract should only report numerical results that are also reported somewhere else in the manuscript. Please check to ensure that all numerical results are also reported somewhere else in the manuscript. For example: "'-0.947' AND '0.013'" does not appear elsewhere in the manuscript.

More Information About This Issue

The abstract must be a summary of the data presented in the manuscript and not a place where unique information is presented. It can be tempting to summarize data differently in the abstract than in the results, but for the sake of clarity this should be avoided. It is a good reporting practice to report the numerical findings in the abstract using the exact same numbers as in the results and tables.

Report Layout: Fundamentals of Reporting

Problem Statement

1

When reporting a proportion in the abstract (e.g., "66%"), it is important to also report the numerator and denominator. For example, 50/100 (50%) patients recovered fully. Providing this information will allow your readers to evaluate the sample size and any missing data.

Explanation

More Information About This Issue

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Examples of how to fix

Where to get more information

Report Layout: Reporting Checklist

Bonus: Checklist Items from the Consort guideline  Reporting Guideline

15

Reporting the nature of the study design (e.g., "Randomized Controlled Trial") directly in the title will help your readers identify the study as an RCT.

16

Please ensure that the following is reported:

- blinding used (e.g., non-blinded, double-blinded)- nature of the design (e.g., parallel-arm, cluster-randomized)- the nature of the intervention (e.g., patients were randomized to receive propranolol or placebo)- the primary outcome of the study (e.g., The primary outcome was 30-day mortality)- the number of patients who were included in the analysis (e.g., 42 patients were included in the analysis...)

 Problem Statement

17

It is important to report the specifics of the trial design, specifically:- The type of trial (e.g., parallel groups, multi-arm, crossover, cluster, factorial) - The nature of the hypotheses (e.g., superiority, equivalence, non-inferiority)- The nature of the randomization ratio (e.g., 1:1)

Types of Errors

- **Errors of Omission** - Things that we could not find in your paper that should be present. By clearly reporting the necessary elements, readers of your work will be able to evaluate what you have done, and even be able to replicate your work. Issues of omission can be fixed by providing the information that is identified in your report.
- **Errors of Commission** - Things that we found in your paper that didn't look right. These issues can represent an error, a misuse of certain terms, or the use of certain procedures that are clearly wrong. These issues are directly listed to you using examples from your actual paper. Issues of commission require some revising of existing descriptions (or changing the underlying methods) to make them right.

Tips & Tricks

- Section Headings!
 - IMRAD
 - Title
 - Introduction
 - Methods
 - Statistical Analyses
 - Results
 - Discussion
 - Table
 - Figure Captions

Tips & Tricks

- Clear methodological design reporting leads to a better report
 - Title: “Lithium in the treatment...: A randomized controlled trial”
 - Abstract: “N = 45 individuals were enrolled in a cross-sectional observational study...”
 - Methods: “Rats were randomized to receive...”

Tips & Tricks

- Many issues are commonly seen
 - When looking at 50 reports, the same item may be seen in 30 of them!
 - Part of the goal of StatReviewer is to encourage consistent use of commonly omitted reporting items
- We suspect that you already know what we have learned:
 - That most of submitted manuscripts contain many of the same errors

Tips & Tricks

- There will be errors!
 - StatReviewer's algorithms are reliable and should focus our attention on issues that deserve attention.
 - However, there will be errors that were triggered by unexpected word choices or styles.
 - Judgment is needed in such cases

Help us do better

- We are constantly improving the algorithms
 - We find errors and fix them
- We are constantly improving the report
 - Are there ways we can improve the report?
 - We would like to make the report most helpful to you!
- New items and functionality are added every week!